R110/B_private_ce | Method-conditioned nodes and actual latency

Complete standalone model including candidate node refinement; unequal parameter counts and no equal-FLOPs claim.

Problem	Instances	Milliseconds / instance
TSP	128	0.990970
CVRP	128	1.457068
ATSP	128	0.142647
OVRP	128	0.152441
VRPB	128	0.110295
VRPL	128	0.151917
VRPTW	128	0.158706
OVRPTW	128	0.156827
OVRPB	128	0.116241
OVRPL	128	0.161228
VRPBL	128	0.115375
VRPBTW	128	0.110309
VRPLTW	128	0.152149
OVRPBL	128	0.110239
OVRPBTW	128	0.110547
OVRPLTW	128	0.151842
VRPBLTW	128	0.110790
OVRPBLTW	128	0.110250

Total trainable parameters: 10,220,553; candidate-conditioned nodes, 21 global-ID-private two-layer graph banks

Optional saved references: R94B=not_requested, R93A=not_requested, R109A=not_requested, R109B=not_requested

R110 refines each candidate node graph through two layers on shared original H4 before pooling. A_tied_ce ties these weights; B_private_ce has 21 banks keyed by global method ID, never candidate column, winner or outcome. Both retain candidate conditioning, original two joint layers and score MLP/maximin. TRAIN optimizes one-forward unweighted legal-pool raw-utility CE, weight 1; comparison, relative-risk and RDrop/KL weights are zero. Evaluation maximin CE is diagnostic, not the optimized raw-utility CE. A has 2,305,033 trainables; B has 10,220,553. The original 1,842,953-parameter base is fresh matched, and all late branches start from identical original-layer prototypes. The initial eval function is identical, but complete state hashes/dimensions are not identical. This is not parameter/FLOP-matched, a routing experiment, or an isolated capacity-only control. Original R88B physical TSP node0 role, methodID+34 and geometry/distance remain; 57,600 updates, logical batch1024, seed2. No new route/path fields, labels, donor, teacher, context, targets, calibration, quotas or auxiliary loss. R89A AdamW, R89B Muon and R84A are saved historical references, not rerun or newly parameter/FLOP-matched controls; R94B, R93A and R109 arms are explicit optional saved references, never implicit dependencies. NSS uses saved historical R79 weights, not retrained on new90k. Fixed36 true rescues/losses, false picks and aligned population net costs/regret are separate from selection frequency. Native/FP64 ind and legal pools; TRAIN-fitted SBS; ALL task-equal. TEST is development-exposed; one whole-model full18 VAL checkpoint per arm for full TRAIN/VAL/TEST/LIB; no TEST/LIB reselection. Actual standalone GPU-resident-input timing includes complete method-conditioned node refinement, excludes dataset I/O, CPU transfer and solvers, and is not end-to-end dataset latency. Microcaps/eval batch come from the main-witness locked runner, not this reporter. No routing summary is required. No online W&B equality/availability is asserted.

R110/B_private_ce | Same selected model: full TRAIN

Diagnostic only; TRAIN was never used to select the checkpoint; full TRAIN-fitted SBS.

Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
ALL	71.2645	66.0332	11.389609	11.400702	11.345073	-0.1936	+0.3913	0.3996
TSP	46.3593	32.8617	8.004967	8.023959	7.970215	-0.2367	+0.4360	0.4324
CVRP	46.3247	18.1284	18.045619	18.151987	17.946816	-0.5860	+0.5505	0.7040
ATSP	76.4716	46.9210	1.640320	1.678679	1.633938	-2.2851	+0.3906	0.3966
OVRP	41.8148	34.2420	7.443742	7.453531	7.397676	-0.1313	+0.6227	0.6349
VRPB	80.6358	80.2765	9.356107	9.357264	9.319342	-0.0124	+0.3945	0.4020
VRPL	37.7654	31.5309	12.393838	12.400882	12.344503	-0.0568	+0.3997	0.4054
VRPTW	80.4531	80.2407	18.812529	18.813703	18.757954	-0.0062	+0.2909	0.3051
OVRPTW	83.9136	83.8025	10.118089	10.118640	10.093220	-0.0054	+0.2464	0.2416
OVRPB	88.4333	88.3580	6.363305	6.363230	6.346658	+0.0012	+0.2623	0.2710
OVRPL	41.5864	34.4716	7.444960	7.453219	7.398483	-0.1108	+0.6282	0.6387
VRPBL	79.8802	79.4593	9.401761	9.403115	9.363196	-0.0144	+0.4119	0.4138
VRPBTW	79.6815	79.4914	19.731766	19.733549	19.642712	-0.0090	+0.4534	0.4376
VRPLTW	80.5296	80.3827	18.768636	18.769959	18.714366	-0.0070	+0.2900	0.3046
OVRPBL	88.6111	88.5556	6.359162	6.358857	6.342513	+0.0048	+0.2625	0.2710
OVRPBTW	83.3457	83.2432	10.623513	10.624446	10.586225	-0.0088	+0.3522	0.3266
OVRPLTW	83.8247	83.7321	10.110642	10.111265	10.085768	-0.0062	+0.2466	0.2435
VRPBLTW	79.7358	79.5630	19.769075	19.770814	19.680364	-0.0088	+0.4508	0.4332
OVRPBLTW	83.3938	83.3370	10.624931	10.625544	10.587365	-0.0058	+0.3548	0.3312

ALL task-equal; ratios of mean costs differ from per-instance actual regret.

R110/B_private_ce | Fixed target36 vs R89A

True rescues/losses, false picks and aligned net cost; fixed targets, no reselection.

Split	Winners	Rescued	Lost	New false	Net cost delta
TRAIN	83279	632	100	788	0.000496406
VAL	4625	30	6	43	0.000733768
TEST	4598	32	4	52	0.000573641

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	0.019959	32359	32068	0.000496406	0.004933
VAL	-0.154321	1693	1818	0.000733768	0.006708
TEST	0.001235	1794	1793	0.000573641	0.005213

Historical only, not retrained/matched; R89A AdamW, R89B Muon, R84A AdamW.

Fixed-target-only deltas do not replace the full18 population net cost/regret above.

R110/B_private_ce | Fixed target36 vs R89B

True rescues/losses, false picks and aligned net cost; fixed targets, no reselection.

Split	Winners	Rescued	Lost	New false	Net cost delta
TRAIN	83279	383	544	625	0.001532732
VAL	4625	14	27	35	0.001271747
TEST	4598	16	17	39	0.000888676

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	-0.426680	26740	32961	0.001532732	0.015999
VAL	-0.351852	1476	1761	0.001271747	0.011713
TEST	-0.186420	1567	1718	0.000888676	0.009803

Historical only, not retrained/matched; R89A AdamW, R89B Muon, R84A AdamW.

Fixed-target-only deltas do not replace the full18 population net cost/regret above.

R110/B_private_ce | Fixed target36 vs R84A

True rescues/losses, false picks and aligned net cost; fixed targets, no reselection.

Split	Winners	Rescued	Lost	New false	Net cost delta
TRAIN	83279	639	73	783	0.000448749
VAL	4625	31	3	46	0.000954554
TEST	4598	33	2	54	0.000516084

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	0.020576	31553	31253	0.000448749	0.004435
VAL	-0.188889	1669	1822	0.000954554	0.008895
TEST	-0.002469	1741	1743	0.000516084	0.004602

Historical only, not retrained/matched; R89A AdamW, R89B Muon, R84A AdamW.

Fixed-target-only deltas do not replace the full18 population net cost/regret above.